

Comparison of Pseudospectral and Spectral Approximation

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The accuracy of pseudospectral (collocation) approximation is compared to spectral (Galerkin) approximation in some simple model problems. It is found that both approximations give similar errors, despite the inclusion of aliasing terms in pseudospectral approximation.

1. Introduction

This paper compares two numerical methods of spectral type for solution of initial-value problems. We restrict attention here to some very simple model problems; comparisons on more realistic problems are presented elsewhere [1]. It is shown that the widely applicable and efficient pseudospectral approximations are roughly as accurate as spectral approximations.

The difference between pseudospectral and spectral approximations is best explained for the simple hyperbolic problem

$$\frac{\partial A}{\partial t} + v(x) \frac{\partial A}{\partial x} = 0, \quad (1)$$

where $A(x, t)$ and $v(x)$ are assumed periodic in x with period 2π .

As proposed in [2, Section 7], pseudospectral approximation to (1) involves the use of a suitable series expansion of $A(x, t)$ to compute accurate values of $\partial A / \partial x$. With periodic boundary conditions, Fourier series are appropriate. The idea is that if

$$A(x) = \sum_{k=-\infty}^{\infty} \bar{a}(k) \exp(ikx),$$

then

$$\frac{\partial A}{\partial x} = \sum_{k=-\infty}^{\infty} ik \bar{a}(k) \exp(ikx).$$

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In fact, if $A(x)$ is specified by its values at the $N = 2K + 1$ equally spaced points $x_n = 2\pi n/N$ ($n = 1, \dots, N$) then $A_n(t) \equiv A(x_n, t)$ is expressible as the discrete Fourier transform

$$A_n = \sum_{|k| \leq K} a(k) \exp(ikx_n) \quad (n = 1, \dots, N), \quad (2)$$

where

$$a(k) = \frac{1}{N} \sum_{n=1}^N A_n \exp(-ikx_n) \quad (|k| \leq K). \quad (3)$$

Also, the relation between $\bar{a}(k)$ and $a(k)$ is

$$a(k) = \bar{a}(k) + \sum_{m \neq 0} \bar{a}(k + mN) \quad (|k| \leq K), \quad (4)$$

where the summation terms are referred to as “aliasing” terms; the “aliases” $k_m = k + mN$ of k satisfy $\exp(ik_m x_n) = \exp(ikx_n)$ so they are indistinguishable from k on the discrete grid x_n .

It follows from (2) that a good approximation to $\partial A/\partial x$ at $x = x_n$ should be

$$\left(\frac{\partial A}{\partial x} \right)_n = \sum_{|k| \leq K} ika(k) \exp(ikx_n) \quad (n = 1, \dots, N). \quad (5)$$

If $A(x)$ is infinitely differentiable and periodic, then $\bar{a}(k)$ tends to zero faster than any power of $1/k$ as $|k| \rightarrow \infty$. Consequently, by (4), the difference between $\bar{a}(k)$ and $a(k)$ tends to zero faster than any power of $1/N$ as $N \rightarrow \infty$, so that the error in (5) also tends to zero faster than any power of $1/N$. Notice also that the discrete transforms (2), (3), (5) are efficiently implementable in order $N \log N$ arithmetic operations by the fast Fourier transform algorithm [3].

A pseudospectral approximation to (1) is

$$\frac{\partial A_n}{\partial t} + v(x_n) \left(\frac{\partial A}{\partial x} \right)_n = 0 \quad (n = 1, \dots, N), \quad (6)$$

where $(\partial A/\partial x)_n$ is given in terms of A_n ($n = 1, \dots, N$) by (3) and (5). If the discrete Fourier coefficients of $v(x_n)$ are $u(k)$ ($|k| \leq K$) by analogy with (2), then the inverse transform (3) of (6) gives the equivalent system

$$\begin{aligned} \frac{\partial a(k)}{\partial t} + \sum_{\substack{p+q=k \\ |p|, |q| \leq K}} iqu(p)a(q) + \sum_{\substack{p+q=k+N \\ |p|, |q| \leq K}} iqu(p)a(q) \\ + \sum_{\substack{p+q=k-N \\ |p|, |q| \leq K}} iqu(p)a(q) = 0 \quad (|k| \leq K). \end{aligned} \quad (7)$$

The last two terms on the left-hand side of (7) are due to aliasing [2, Section 3].

In the form (6), pseudospectral approximation is equivalent to collocation, i.e. the truncation error in approximating (1) is formally set equal to zero at the collocation points x_n . In the form (7), pseudospectral approximation is equivalent to including aliasing terms at full strength, but evaluating derivatives without phase error [4, Section 4(i)]. The latter property is characteristic of all approximations of spectral type and distinguishes such approximations from finite-difference or finite-element approximations.

On the other hand, spectral (Galerkin) approximation drops the aliasing terms in (7), i.e.

$$\frac{\partial a(k)}{\partial t} + \sum_{\substack{p+q=k \\ |p|, |q| \leq K}} iqu(p)a(q) = 0 \quad (|k| \leq K). \quad (8)$$

In effect, the Galerkin approximation (8) formally enforces the condition that the error in approximating (1) be orthogonal to the expansion functions $\exp(ikx)$ ($|k| \leq K$) [2, Section 2]. Some reasons for believing that aliasing terms are errors are given in [4, Section 4(v)]. However, the present analysis will demonstrate that aliasing is not as evil as usually thought.

Pseudospectral approximation possesses several operational advantages over spectral approximation. First, solution of (6) or (7) requires only two fast Fourier transforms per time step, viz. (3) and (5) to evaluate (6), or (2) and the inverse transform of $v(x_n)(\partial A/\partial x)_n$ to evaluate (7) by a convolution theorem. On the other hand, evaluation of the convolution in (8) requires four fast Fourier transforms per time step [2, Section 3]. For more complicated problems in several space dimensions, the fastest spectral method [2, Section 6] requires at least twice the number of fast Fourier transforms as the corresponding pseudospectral method [2, Section 7]. Also, pseudospectral codes are considerably simpler than spectral codes. Finally, pseudospectral approximation applies to a much more general class of problems than spectral approximation [2, Section 7].

The single disadvantage of pseudospectral approximation is that it may be susceptible to numerical instability due to the aliasing terms [5]. However, most realistic physical problems have some kind of damping associated with them that limit aliasing and, along with it, aliasing instability (cf. [1]).

2. Nonuniform advection

We consider the model problem (1) with $v(x)$ one-signed, say positive, in order to avoid the development of singularities of $A(x, t)$ in time. A stagnation point of $v(x)$ invalidates the following analysis (cf. [4, Section 5]). The overall accuracy of the approximations (6) or (8) to (1) is best assessed in terms of the eigenfunctions of the spatial part of (1):

$$\psi_p(x) = \exp\left[ipv \int_{-\pi}^x ds/v(s)\right], \quad (9)$$

where p is integral and

$$v \int_{-\pi}^{\pi} ds/v(s) = 2\pi. \quad (10)$$

The general solution to (1) may be expanded as

$$A(x, t) = \sum_{p=-\infty}^{\infty} \alpha_p \psi_p(x) \exp(-ipvt). \quad (11)$$

Because of (11), the faithfulness of the eigenvalues and eigenfunctions of the spatial parts of (6) and (8) to the exact eigenvalues ipv and eigenfunctions $\psi_p(x)$ is a measure of the overall accuracy of simulation.

If $v(x)$ is constant, the spectra of (6) and (8) are identical and are a subset of the exact spectrum: the eigenvalues are ipv for $|p| \leq K$. If $v(x)$ is not constant, neither (6) or (8) is generally exact. However, if $v(x)$ is one-signed, the eigenvalues of (6) and (8) are pure imaginary and occur as complex conjugate pairs, except for the eigenvalue 0. We show that the pseudospectral approximation (6) has only pure imaginary eigenvalues by setting $A(x, t) = A(x) \exp(\lambda t)$ so that

$$\lambda \sum_{n=1}^N \frac{|A_n|^2}{v_n} = \sum_{n=1}^N A_n^* \left(\frac{\partial A}{\partial x} \right)_n, \quad = - \sum_{n=1}^N \left(\frac{\partial A}{\partial x} \right)_n^* A_n, \quad = -\lambda^* \sum_{n=1}^N \frac{|A_n|^2}{v_n}.$$

It follows that $\text{Re}(\lambda) = 0$. The second of the above equalities follows easily from (3) and (5). On the other hand, we have not been able to prove that $\text{Re}(\lambda) = 0$ for (8) when $v(x) > 0$, although the result is established generally for $K \leq 3$ and for a wide range of special cases for $K \leq 30$.

Table 1
Eigenvalues of $v(x)\partial/\partial x$ for $v(x) = 1/(1 + \frac{1}{2} \sin x)^1$

Exact	Pseudospectral			Spectral		
	$K = 3$	$K = 4$	$K = 5$	$K = 3$	$K = 4$	$K = 5$
1	1.00003	1.00000	1.00000	1.00004	1.00000	1.00000
2	2.03329	2.00158	2.00004	2.04652	2.00260	2.00007
3	3.76124	3.12367	3.01285	3.87844	3.15524	3.01803

¹ Only the imaginary parts of the eigenvalues with the smallest positive imaginary parts are listed; the real parts are zero. Also, there exists a zero eigenvalue and eigenvalues with imaginary parts equal to the negative of those listed. In Table 1, the value of v [cf. (10)] is 1.

The spectra of (6) and (8) have been determined numerically using the *QR* matrix eigenvalue algorithm. Some results are given in Tables 1–3. Only the imaginary parts of the eigenvalues with the smallest positive imaginary parts are listed. It is apparent from Tables 1 and 2 that if $v(x)$ is infinitely differentiable,

Table 2
Eigenvalues of $v(x)\partial/\partial x$ for $v(x) = 1/(1 + \frac{1}{2} \sin 2x + \frac{1}{4} \sin 3x + \frac{1}{10} \sin 5x)^2$

Exact	Pseudospectral			Spectral		
	$K = 5$	$K = 7$	$K = 9$	$K = 5$	$K = 7$	$K = 9$
1	1.00110	1.00001	1.00000	0.96777	0.99205	0.99957
2	2.00908	2.00031	2.00004	1.95564	1.98574	1.99934
3	3.06880	3.01849	3.00200	3.04688	3.01916	3.00435
4	4.87225	4.10320	4.01140	5.08852	4.14905	4.02230
5	7.07687	5.23210	5.08486	7.90377	5.29910	5.11704

² See the footnote to Table 1. Here $v = 1$.

Table 3
Eigenvalues of $v(x)\partial/\partial x$ for $v(x) = 1 + |\sin x|^3$

Exact	Pseudospectral			Spectral		
	$K = 7$	$K = 10$	$K = 16$	$K = 7$	$K = 10$	$K = 16$
1.57080	1.56536	1.56795	1.56962	1.56654	1.56839	1.56971
3.14159	3.13393	3.13624	3.13927	3.13831	3.13718	3.13947
4.71239	4.70635	4.70654	4.70907	4.71271	4.70936	4.70952
6.28319	6.33081	6.27806	6.27898	6.35485	6.28199	6.27962
7.85398	7.94704	7.87101	7.84981	7.97382	7.88376	7.85125
9.42478	10.03543	9.46040	9.42071	10.11494	9.47567	9.42258
10.99557	11.80244	11.21416	10.99737	11.87648	11.25832	11.00211

³ See the footnote to Table 1. Here $v = \frac{1}{2}\pi$.

then the eigenvalues of (6) and (8) converge with great rapidity. On the other hand, the eigenvalues when $v(x) = 1 + |\sin x|$ given in Table 3 converge only like $1/K^2$ because $v'(x)$ is discontinuous at $x = 0, \pi, 2\pi$. The results given in Tables 1–3 show that the pseudospectral approximation (6) is at least as accurate as the spectral approximation (8), despite aliasing “errors.”

3. Rigid boundary conditions

Periodic boundary conditions are *not* essential to the pseudospectral method. However, other boundary conditions require different series expansions in order to ensure rapid convergence. A simple example is given by the one-dimensional wave equation

$$\frac{\partial A}{\partial t} + \frac{\partial A}{\partial x} = 0 \quad (|x| \leq 1), \quad (12a)$$

$$A(-1, t) = f(t), \quad (12b)$$

for which Chebyshev series are appropriate [6].

Chebyshev pseudospectral approximation to (12) is developed as follows. We expand $A(x)$ as

$$A(x) = \sum_{p=0}^N a_p T_p(x), \quad (13)$$

where $T_p(x)$ is the Chebyshev polynomial of degree p defined by

$$T_p(\cos \theta) = \cos p\theta.$$

The $N + 1$ Chebyshev coefficients a_p are chosen so that (13) gives $A_n \equiv A(x_n)$ at the $N + 1$ collocation points $x_n = \cos \pi n/N$ ($0 \leq n \leq N$), i.e.

$$A_n = \sum_{p=0}^N a_p \cos \pi p n/N \quad (0 \leq n \leq N). \quad (14)$$

The inverse of (14) is

$$\bar{c}_p a_p = \frac{2}{N} \sum_{n=0}^N \bar{c}_n^{-1} A_n \cos \pi p n / N \quad (0 \leq p \leq N), \quad (15)$$

where $\bar{c}_0 = \bar{c}_N = 2$, $\bar{c}_p = 1$ for $0 < p < N$. The important fact is that the discrete Fourier transforms (14), (15) are efficiently computable in order $N \log N$ operations by the fast Fourier transform algorithm.

According to (13),

$$A'(x) = \sum_{p=0}^N a_p T'_p(x), \quad = \sum_{p=0}^N a_p^{(1)} T_p(x) \quad (16)$$

where

$$a_p^{(1)} = \frac{2}{c_p} \sum_{\substack{n=p+1 \\ n+p \text{ odd}}}^N n a_n \quad (17)$$

and $c_0 = 2$, $c_p = 1$ for $p > 0$. The results (16), (17) follow from the properties of Chebyshev polynomials [6]. In pseudospectral approximation, we compute $A'(x_n)$ by

$$A'_n \equiv A'(x_n) = \sum_{p=0}^N a_p^{(1)} \cos \pi p n / N \quad (0 \leq n \leq N). \quad (18)$$

Since the set of $a_p^{(1)}$ is calculable in terms of the set of a_p using (17) in order N operations, and (15) and (18) are implementable using fast Fourier transforms, the calculation of A'_n from A_n requires only order $N \log N$ operations.

The pseudospectral approximation to (12) is

$$\frac{\partial A_n}{\partial t} = -A'_n \quad (0 \leq n < N), \quad (19a)$$

$$A_N = f(t). \quad (19b)$$

These equations may be combined into

$$\frac{\partial A_n}{\partial t} = -A'_n + N b(t) \delta_{nN} \quad (0 \leq n \leq N), \quad (20)$$

where $\delta_{nN} = 1$ if $n = N$, 0 otherwise, and $N b(t) = f'(t) + A'_N$. If the inverse transformation (15) is applied to (20), it follows that pseudospectral approximation gives

$$\frac{\partial a_p}{\partial t} = -a_p^{(1)} + \bar{c}_p^{-1} (-1)^p b(t) \quad (0 \leq p \leq N), \quad (21a)$$

$$\sum_{p=0}^N (-1)^p a_p = f(t). \quad (21b)$$

Equations (21) give $N + 2$ equations in the $N + 2$ dependent variables $a_p(t)$ ($0 \leq p \leq N$), $b(t)$.

On the other hand, the Galerkin (spectral) approximation to (12) using Chebyshev polynomial expansions is [6]

$$\frac{\partial a_p}{\partial t} = -a_p^{(1)} + c_p^{-1}(-1)^p b(t) \quad (0 \leq p \leq N), \quad (22a)$$

$$\sum_{p=0}^N (-1)^p a_p = f(t). \quad (22b)$$

The only difference between (21) and (22) is the coefficient of $b(t)$. For $p = N$, $b(t)$ is multiplied by $\frac{1}{2}$ in (21a) and by 1 in (22a).

It was shown in [6] that if $f(t)$ is infinitely differentiable then the solutions to (22) converge faster than any power of $1/N$ to solutions of (12). The same proof extends easily to (21).

In practice, the errors involved in (21) and (22) are quite similar. For example, with $f(t) = \sin 3\pi t$, the rms errors of the spectral approximation (22) are roughly 1.5×10^{-2} with $N = 13$ and 2.1×10^{-3} with $N = 15$, while the errors of the pseudospectral approximation (21) are 3.3×10^{-2} with $N = 13$ and 3.8×10^{-3} with $N = 15$.

The very simple relation between the pseudospectral and spectral approximations (21) and (22), respectively, seems to be true only for constant coefficient problems (like the heat equation). In more complicated problems, pseudospectral approximations include aliasing terms while spectral approximations do not.

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References

1. FOX, D. G. AND ORSZAG, S. A., "Pseudospectral approximation to two-dimensional turbulence," to be published (1972).
2. ORSZAG, S. A., "Numerical simulation of incompressible flows within simple boundaries. I. Galerkin (spectral) representations," *Stud. in Appl. Math.* **50**, 293-327 (1971).
3. COOLEY, J. W. AND TUKEY, J. W., "An algorithm for the machine computation of complex Fourier series," *Math. Comp.* **19**, 297-301 (1965).
4. ORSZAG, S. A., "Numerical simulation of incompressible flows within simple boundaries: accuracy," *J. Fluid Mech.* **49**, 75-112 (1971).
5. PHILLIPS, N. A., "An example of nonlinear computational instability," in *The Atmosphere and the Sea in Motion*, pp. 501-504, Rockefeller Institute Press (1959).
6. ORSZAG, S. A., "Galerkin approximations to flows within slabs, spheres, and cylinders," *Phys. Rev. Letters* **26**, 1100-1103 (1971).

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